



Department of Computer Technology

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027(ODD)

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Tanushrif

Name and Signature of Student and Date
(Signature and Date in Handwritten)



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Session	2026-27 (ODD)	Course Name	LAB: Software Engineering
Semester	7 th	Course Code	23CT1704
Roll No	25	Name of Student	Tanushri Sawalikar

Practical Number	08
Aim	To write a program to estimate effort, development time, staffing, and cost of a software project using the Basic COCOMO model.
Problem Definition	Software project estimation is important for determining the resources, time, effort, and cost required to develop a software system. The COCOMO model is used to estimate the effort and development time based on the size of the software measured in KLOC (Kilo Lines of Code). The project can be classified into Organic, Semi-Detached, or Embedded mode. The program accepts the project size, project mode, and cost per person-month and calculates the estimated effort, development time, average staff, and total project cost.
Theory (100 words)	COCOMO (Constructive Cost Model) is a software cost estimation model developed by Barry Boehm. It estimates the effort and development time required for a software project based mainly on its size in KLOC. Basic COCOMO classifies projects into three modes: Organic, Semi-Detached, and Embedded. Effort is calculated in person-months using $\text{Effort} = a \times (\text{KLOC})^b$, while development time is calculated using $\text{Time} = c \times (\text{Effort})^d$. Average staffing is obtained by dividing effort by development time. The total cost can be calculated by multiplying effort by the cost per person-month.
Procedure and Execution (100 Words)	Algorithm(Steps): Step 1: Start. Step 2: Read project size KLOC. Step 3: Select project mode. Step 4: Assign a, b, c, and d according to the selected mode. Step 5: Read cost per person-month. Step 6: Calculate effort: $\text{Effort} = a \times (\text{KLOC})^b$ Step 7: Calculate development time: $\text{Time} = c \times (\text{Effort})^d$



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Step 8: Calculate average staff:

Staff = Effort / Time

Step 9: Calculate cost:

Cost = Effort × Cost per Person-Month

Step 10: Display effort, development time, average staff, and cost.

Step 11: Stop.

Code:

```
#include <stdio.h>
#include <math.h>
int main()
{
    float kloc, costPerPM;
    float a, b, c, d;
    float effort, time, staff, cost;
    int mode;
    printf("COCOMO Software Cost Estimation\n");
    printf("-----\n");
    printf("Enter project size in KLOC: ");
    scanf("%f", &kloc);
    printf("\nSelect Project Mode:\n");
    printf("1. Organic\n");
    printf("2. Semi-Detached\n");
    printf("3. Embedded\n");
    printf("Enter your choice: ");
    scanf("%d", &mode);
    switch(mode)
    {
        case 1:
            a = 2.4;
            b = 1.05;
            c = 2.5;
            d = 0.38;
            break;
        case 2:
            a = 3.0;
            b = 1.12;
            c = 2.5;
            d = 0.35;
            break;
        case 3:
```

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```
a = 3.6;
b = 1.20;
c = 2.5;
d = 0.32;
break;
default:
printf("Invalid project mode!\n");
return 0;
}
printf("Enter cost per person-month: ");
scanf("%f", &costPerPM);
effort = a * pow(kloc, b);
time = c * pow(effort, d);
staff = effort / time;
cost = effort * costPerPM;
printf("\n----- COCOMO ESTIMATION ----- \n");
printf("Project Size      : %.2f KLOC\n", kloc);
printf("Effort             : %.2f Person-Months\n", effort);
printf("Development Time    : %.2f Months\n", time);
printf("Average Staff       : %.2f Persons\n", staff);
printf("Estimated Cost      : %.2f\n", cost);
return 0;
}
```

Output:

```
C:\Users\student\Documents\ X + v
COCOMO Software Cost Estimation
-----
Enter project size in KLOC: 1

Select Project Mode:
1. Organic
2. Semi-Detached
3. Embedded
Enter your choice: 1
Enter cost per person-month: 15000

----- COCOMO ESTIMATION -----
Project Size      : 1.00 KLOC
Effort           : 2.40 Person-Months
Development Time  : 3.49 Months
Average Staff    : 0.69 Persons
Estimated Cost   : 36000.00

Process returned 0 (0x0)   execution time : 28.690 s
Press any key to continue.
```

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Output Analysis	For a project size of 10 KLOC using the Organic COCOMO model , the program calculates the required software development effort, development time, average staffing, and estimated cost. The calculated effort represents the total person-months required for development. Development time represents the estimated duration of the project, while average staff represents the approximate number of people required during development. The final cost is calculated using the effort and cost per person-month.
Link of student Github profile where lab assignment has been uploaded	https://github.com/TanushriSawalikar/Software-Engineering-Practicals
Conclusion	The program successfully implements the Basic COCOMO model to estimate software project effort, development time, average staffing, and cost. The estimation changes according to the project size and selected project mode. Thus, COCOMO can be used as a basic technique for software project planning and cost estimation.
Plag Report (Similarity index < 12%)	The screenshot shows a plagiarism checker interface with tabs for 'Plagiarism Checker', 'Check Grammar', and 'AI Detector'. The 'Plagiarism Checker' tab is active. On the left, 'Scan Properties' are listed: Sources Found (0), Words (208), and Characters (2019). On the right, a large circular gauge shows 'Plagiarism 0%' and 'Unique 100%'. Below the gauge, it indicates '0% Exact Match' and '0% Partial Match'. A 'View More Details' button is visible at the bottom left of the scan area.
Date	28/09/26